

Toward SDN-Controlled Microgrid Reconfiguration: An Auditable Python–PSCAD Switching Testbed

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Abstract

Separating network-control development from electrical-model development is useful for studying programmable feeder restoration, but a graph-valid route alone does not establish its electrical behavior. This paper presents an SDN-inspired Python–PSCAD testbed that translates structured controller commands into timed low-voltage breaker operations, executes an electromagnetic transient experiment, and returns traceable measurements to a graphical dashboard. The implementation retains the networking simulator separately from the power-plane integration and preserves JSON as the exchange contract beneath graphical editing. A six-bus, seven-branch, 400-V feeder with two normally open ties provides the initial electrical case. Six freshly rerun PSCAD experiments examine baseline operation, isolation, restoration, loop rejection, imposed control delay, and a native three-phase-to-ground fault. In the physical-fault case, the maximum absolute phase current is 1.554 kA; healthy downstream buses recover to 380.5–387.4 V after approximately 1.195 s below a 200-V analysis threshold, while the faulted bus remains isolated. An additional replay of fault events from the merged networking simulator schedules electrical isolation for a power fault and no breaker action for a communication fault. The contribution is an auditable command-to-EMT workflow and its preliminary functional evidence, not a new optimal reconfiguration algorithm or a demonstration of real-time closed-loop SDN protection. Illustrative electrical parameters, non-transactional command handling, and the absence of live controller feedback delimit the results and motivate the next validation stage.

Keywords: software-defined networking; low-voltage feeder; microgrid reconfiguration; PSCAD; electromagnetic transients; reproducibility.

1 Introduction

Distribution feeder reconfiguration changes which sectionalizing and tie switches conduct, allowing the same physical infrastructure to supply loads through different connected configurations. Classical work treats this as an electrical-network problem with objectives such as loss reduction and load balancing [1]. Software-defined networking (SDN), by comparison, separates programmable forwarding decisions from the mechanisms that execute them in a communication network [2]. The conceptual separation is attractive for a project in which one team develops network-control logic and another investigates physical low-voltage (LV) switching.

The analogy has an important boundary. Alternating-current power is not forwarded as independently addressed packets through passive feeder branches. Switching alters circuit connectivity, after which voltages and currents follow the network impedances, source characteristics, and electrical dynamics. Consequently, a route returned by a networking algorithm must be translated into a physically represented switching sequence and evaluated in an electrical simulator. A graph can establish connectivity; it cannot, by itself, establish acceptable voltage, equipment loading, transient behavior, or protection coordination.

The broader project aims to maintain a communication control plane independently of switchable power paths. Its present electrical realization is a single-source LV feeder, not a complete multi-resource microgrid. This distinction matters: the U.S. Department of Energy’s microgrid description includes local energy resources, controllability as an entity, and grid-connected or islanded operation [3]. The current case has neither distributed energy resource (DER) control nor a demonstrated islanding transition.

The term “toward” in the title therefore denotes a research trajectory rather than an already validated operating capability.

This paper asks three practical questions. First, can externally specified network commands be mapped consistently onto physical breaker schedules while retaining identifiers and operation-level decisions? Second, do the saved EMT measurements show the expected distinction between fault inception, isolation, and restoration of healthy loads? Third, can graphical dashboard edits trigger fresh, inspectable PSCAD experiments without changing the networking team’s implementation?

The resulting contributions are:

1. A modular command-to-power-plane workflow that retains a structured controller contract, operation-level graph checks, explicit component bindings, and a reproducible timed experiment manifest.
2. A PSCAD automation path with native breaker and fault components, serialized run ownership, fresh-result checks, and a defined set of 31 research channels exposed through a graphical JSON-backed dashboard.
3. An evidence-bounded analysis of six saved deterministic scenarios, including a physical fault, downstream restoration, and an explicitly imposed restoration delay, accompanied by reproducible publication assets.

These are integration and functional-validation contributions. No superiority over established reconfiguration algorithms or cyber-physical testbeds is claimed. The experiments support further research on the interface between network decisions and electrical consequences; they are not evidence that the proposed architecture is ready to operate a physical feeder.

2 Background and Related Work

2.1 Electrical Reconfiguration and Software-Defined Control

Baran and Wu formulated distribution reconfiguration in terms of electrical losses and load balance, establishing the relevance of switch state and radial network structure [1]. The present work uses radiality only as an operation-level topological constraint. It does not implement their optimization procedure, solve an optimal power flow, or benchmark a new route-selection method.

OpenFlow provides an influential example of a programmable interface between network control and forwarding elements [2]. Applying this idea to an energy system requires care over which plane is being programmed. In conventional SDN, the data plane forwards communication traffic. In the project terminology, the “power plane” instead denotes conducting electrical paths; the interface described here is not an OpenFlow implementation for breakers.

Software-defined microgrid control is already an established research topic. Wang et al. virtualize microgrid control functions as software services, reducing dependence on dedicated controller hardware [4]. Li and Du employ SDN and OpenFlow to support programmable communications for networked microgrid formation, partition, and reconfiguration [5]. These works preclude a broad novelty claim that separating control from physical microgrid operation is itself new. The narrower focus here is retaining an evolving network-team command contract while producing an inspectable PSCAD switching experiment and measurement record.

2.2 Communications and Cyber-Physical Simulation

Communication behavior can influence control outcomes, but a delayed timestamp is not equivalent to a simulated communication network. Dorsch et al. compare legacy and software-defined communications in distributed smart-grid control, explicitly considering communication infrastructure and agent interaction [6]. The long-delay case in this paper instead imposes a later restoration time. It illustrates the electrical cost of waiting, without measuring packet loss, routing convergence, or SDN latency.

Federated simulators address a different problem from batch automation. FNCS couples power-system and communication simulators with synchronization strategies for their exchanges [7]. HELICS supports coordinated event-driven and time-series simulations, including iteration among federates [8]. Neither framework is integrated into the current implementation. Their relevance is methodological: a persistent

Table 1: Positioning against selected related work. Entries describe scope, not a ranking or a feature-completeness assessment.

Work	Principal emphasis	Distinction of this study
Baran–Wu [1]	Electrical reconfiguration objectives	Evaluates supplied routes; no optimizer
Wang et al. [4]	Virtualized microgrid controllers	Command-to-EMT experiment integration
Li–Du [5]	SDN-enabled networked microgrids	JSON batch interface, not live OpenFlow
FNCS / HELICS [7, 8]	Synchronized simulator federation	Sequential automation of a single EMT run
Haque et al. [9]	Cyber-physical threat emulation	Illustrative switching and fault scenarios

application connection does not supply the shared-time and causality mechanisms required for closed-loop co-simulation.

More recent cyber-physical emulation also extends well beyond scheduled switching. Haque et al. combine an operational-microgrid-derived power model with a synthetic communication network, DNP3 exchanges, and structured threat scenarios [9]. Their work provides a useful contrast to the illustrative feeder and deterministic commands used here; this paper makes no comparative cybersecurity or resilience-performance claim.

2.3 PSCAD as the Electrical Execution Layer

PSCAD exposes Python automation for loading and running projects [10]. Its timed breaker component supports one or two operations associated with a breaker control signal [11]. The supplied fault models represent a fault using switched resistances rather than a detailed arc characteristic [12]. These documented facilities are sufficient for the deterministic switching cases examined here. The project-specific contribution lies in translating commands, managing run completion and result freshness, and connecting the results to research artifacts; it is not a replacement for the underlying EMT solver.

3 System Architecture and Command Semantics

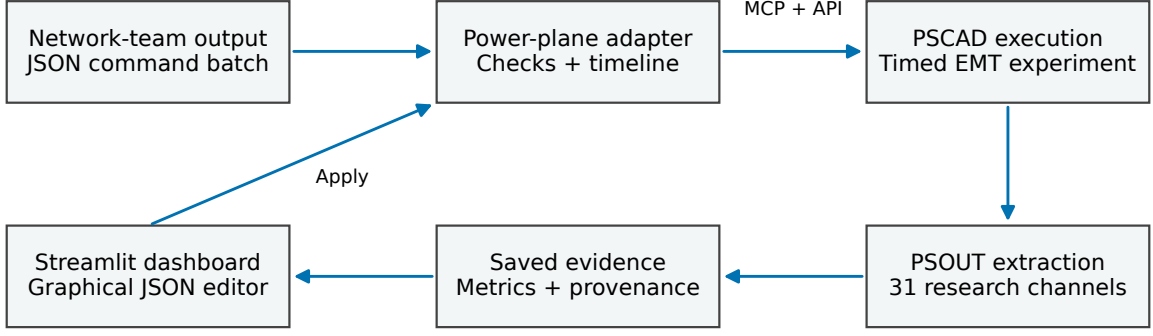
3.1 Ownership Boundaries and Data Flow

The networking simulator remains in its existing repository modules. The power-plane extension is housed separately under `research_power_plane`, with dedicated command adaptation, topology, timeline, PSCAD mapping, automation, measurement, and dashboard modules. This separation allows networking colleagues to evolve their algorithms while the adapter handles changes in their output fields. It does not guarantee compatibility with an arbitrary future schema: new aliases and semantic changes must be tested at the adapter boundary.

The integrated 25 September revision adds packet-driven decentralized route learning, binary packet encoding, selectable LoRa airtime, a shared event-based radio channel, and explicit communication versus electrical fault domains. The current PSCAD interface consumes fault-event rows or explicit command batches; it does not synchronize the radio event clock with the EMT solver. For domain-aware fault rows, the adapter emits an electrical action only for a power fault whose recommendation calls for switching. Communication and device faults do not schedule breakers. Legacy rows without a domain retain their earlier isolation behavior. The merged networking simulator’s `export_tables_json()` omits fault events, so the adapter receives its `fault_events` collection explicitly.

Figure 1 shows the implemented workflow. A controller output batch or a graphical dashboard submission supplies topology, commands, optional faults, and logical-to-PSCAD bindings. Canonical JSON remains the exchange representation. The dashboard provides feeder selection, switch-state controls, scheduled operations, fault forms, and mapping tables; advanced raw JSON remains available. Committing an edit can automatically invoke the same pipeline as a command-line replay. Selecting an element or moving the scheduled-state preview is not a simulation request.

The displayed network state and the PSCAD wiring are related but not interchangeable. The present dashboard operates on the fixed six-bus electrical case and its component map. A new graph edge without a corresponding wired PSCAD branch does not create a new physical circuit. Extending the feeder requires an electrical-model and mapping update.



Each submitted change starts a new batch run; no live solver feedback to the network controller.

Figure 1: Implemented batch workflow. Solid arrows indicate current processing paths. Measurement display follows run completion; no feedback path from the running EMT solver to the networking controller is implemented.

3.2 Command Contract and Graph Model

Represent a normalized command as

$$c_k = (\text{id}_k, \tau_k, a_k, \mathcal{O}_k, \mathcal{M}_k), \quad (1)$$

where τ_k is a simulation-time timestamp, a_k is an action, $\mathcal{O}_k$ contains target nodes, edges, or a requested path, and $\mathcal{M}_k$ holds controller identity and metadata. Supported actions include opening or closing a line, isolating or restoring a node, applying a switch set, and requesting a reroute path. Commands are processed in a stable chronological order, retaining input order for equal timestamps.

Let the physical connectivity model be the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $x_e = 1$ for a logically closed branch and $x_e = 0$ for an open branch. The currently conducting graph is

$$\mathcal{G}_x = (\mathcal{V}, \{e \in \mathcal{E} : x_e = 1\}). \quad (2)$$

For a source set $\mathcal{S}$, the software’s energized-node estimate is the union of connected components containing a source. This is a connectivity estimate, not a voltage calculation. Likewise, the analysis field called “island count” counts connected components; it does not establish viable grid-forming operation within a separated component.

With radiality preservation enabled and an initially acyclic graph, a proposed closure $e = (u, v)$ is rejected if a path already connects u and v in $\mathcal{G}_x$. Opening an edge cannot introduce a cycle. This local rule preserves a forest during accepted edge changes, but does not require every load to remain connected. It also does not enforce voltage limits, branch ratings, source capacity, protection selectivity, or synchronization between multiple sources.

An implementation detail limits the strength of the validation guarantee. The current command handler records command-level warnings while processing individual requested operations. A command may therefore be marked unaccepted or partially accepted after some valid edge operations have already been applied. Only accepted state-changing operations enter the switching timeline; rejected operations and no-ops do not. This is *not an atomic, all-or-nothing transaction*. The loop-rejection experiment tests a single rejected closure, not general rollback behavior. A production interlock would require validation of the complete prospective transition before committing it.

3.3 Schedule Compilation and Execution

Each logical line identifier maps to a PSCAD timed-logic component and its associated three-phase breaker. The component’s control convention reverses the graph variable:

$$b_e(t) = 1 - x_e(t), \quad b_e = 0 \text{ closed}, \quad b_e = 1 \text{ open}. \quad (3)$$

The adapter writes the initial state and operation count and times using `INIT`, `NUMS`, `T01`, and `T02`. The current mapping permits at most two transitions per breaker during one run, consistent with the selected

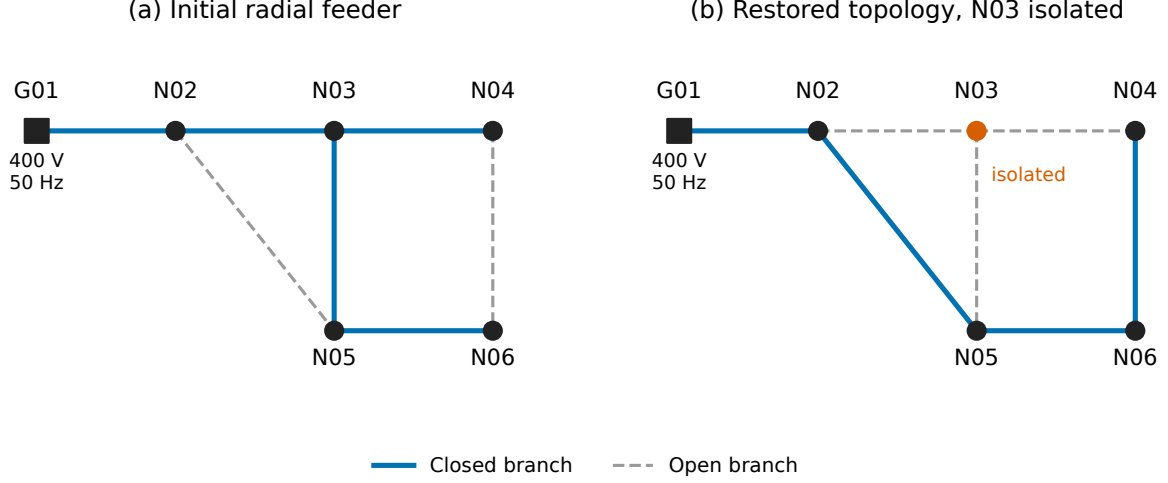


Figure 2: Original schematic of the modeled electrical connectivity before and after restoration. Lines represent three-phase branches, not communication links or geographical cable routing. Node positions are diagrammatic.

timed component [11]. An unchanged branch receives an operation beyond the simulation window to accommodate the component’s lack of a zero-operation setting. The generated waveform inside the experiment window remains constant.

Execution uses a project-local wrapper around PowerMCP, a Model Context Protocol (MCP) server exposing PSCAD automation tools. A persistent localhost service owns the application connection. The local wrapper focuses the intended case and uses a blocking project run so that returning control means the run has completed. The persistence avoids reopening the application for every dashboard action; it does not represent an independently powered, always-available field control network.

The client serializes project parameter writes with a run lock. To avoid the PSCAD output-overwrite dialog, it removes only the current case’s generated output before building. Other case files and separately named archives are preserved. Successful execution requires acceptable build messages, a newly produced output fingerprint, and the expected measured channels. Tool errors, missing channels, or stale results fail the request rather than being presented as a successful new simulation.

These are engineering safeguards, not a formal proof of end-to-end correctness. In particular, clearing the current raw output means that publication-grade reproducibility needs a separate immutable archive for every run. Each new submission starts another continuous EMT experiment from its specified initial conditions. The current system does not alter a running simulation in response to newly arriving controller messages.

4 Electrical Test System

4.1 Feeder and Load Representation

The generated case, `SEIAN_LV_Timed_Switching`, comprises buses G01 and N02–N06, seven three-phase switchable series R/L branches, and five balanced Y-connected resistive loads. G01 is supplied by a Thevenin source configured at 0.4 kV line-to-line RMS and 50 Hz, with a source resistance setting of $0.02\ \Omega$. The declared source setting and implemented branch values are illustrative; they are not a short-circuit calibration against a measured transformer or utility supply.

Five branches form the initial radial tree; N02–N05 and N04–N06 are normally open ties. Opening N02–N03, N03–N04, and N03–N05 isolates N03. Closing both ties then supplies the healthy downstream buses through G01–N02–N05–N06–N04, as shown in Figure 2.

The builder assigns resistance from a nominal branch-capacity field. For a numerical capacity entry q_e in kW, its heuristic is

$$R_e = \frac{3}{q_e}\ \Omega, \quad L_e = \frac{0.3R_e}{2\pi f}, \quad f = 50\ \text{Hz}. \quad (4)$$

The constant 3 is an implementation sizing heuristic with the implied unit conversion, not a conductor

Table 2: Implemented per-phase branch parameters, extracted from the saved PSCAD build report. These values are illustrative, not measured cable data.

Branch	R_e (Ω)	L_e (μH)	Initial state
G01–N02	0.05000	47.75	Closed
N02–N03	0.07500	71.62	Closed
N03–N04	0.08571	81.85	Closed
N05–N06	0.10000	95.49	Closed
N03–N05	0.10000	95.49	Closed
N02–N05	0.12000	114.59	Open (tie)
N04–N06	0.12000	114.59	Open (tie)

Table 3: Resistive load sizing. The demand column refers to 400-V operation, not the measured demand at the loaded bus voltage.

Bus	Sizing label (kW)	Demand at 400 V (kW)	$R_{i,\phi}$ (Ω)
N02	8	7.84	20.40816
N03	8	7.84	20.40816
N04	10	9.80	16.32653
N05	8	7.84	20.40816
N06	6	5.88	27.21088

law. Consequently, `capacity_kw` should not be interpreted as an enforced thermal rating. Table 2 reports the actual generated impedances directly.

The loads require similarly careful interpretation. The source builder contains a scalar named `LOAD_PF`, set to 0.98, but uses it to multiply the target active power when calculating resistance. It does not introduce a reactive load or enforce a 0.98 electrical power factor. If P_i^{set} is expressed in watts and $V_{LL,0} = 400$ V, the implemented phase resistance is

$$R_{i,\phi} = \frac{(V_{LL,0}/\sqrt{3})^2}{0.98P_i^{\text{set}}/3} = \frac{V_{LL,0}^2}{0.98P_i^{\text{set}}}. \quad (5)$$

Thus the nominal-voltage resistive demand is $0.98P_i^{\text{set}}$ and varies approximately with the square of applied balanced RMS voltage. These are not constant-power loads. Table 3 records both the original sizing labels and the resulting nominal-voltage demands.

4.2 Fault and Numerical Configuration

The physical-fault scenario uses a native three-phase-to-ground fault at N03, with an on-resistance of $0.05\ \Omega$ and an off-resistance of $10^6\ \Omega$. Its programmed inception is 4.8 s, duration 0.4 s, and clearing time 5.2 s. All incident N03 breakers are commanded open at 5.0 s and both ties close at 6.0 s. Fault activation uses the native `tpflt` and `tfaultn` components; it is not synthesized by manually changing a plotted voltage trace. Finite off-resistance produces small leakage when the fault is inactive. The native resistive-switch abstraction does not model fault-arc dynamics [12].

PSCAD advances the EMT solution with a $25\ \mu\text{s}$ time step and records research signals every $50\ \mu\text{s}$. A duration- T experiment therefore contains $T/(50\ \mu\text{s}) + 1$ recorded samples per channel when both endpoints are included. Startup is included in the saved trajectories but excluded from the interruption metric as described below. Time-step convergence, frequency dependence of line parameters, grounding calibration, unbalanced demand, inverter dynamics, and transformer detail have not been established for this case.

Table 4: Saved experiment matrix. Accepted commands are shown as accepted/issued. Events count individual breaker transitions. The N04 duration is time below 200 V; $\geq$ marks an interval still ongoing when the run ended.

ID	Condition	Run (s)	Cmd.	Events	Samples/ch.	N04 (s)
S1	Baseline	1	0/0	0	20,001	0.0000
S2	Isolation only	6	1/1	3	120,001	≥ 0.9835
S3	Tie restoration	7	2/2	5	140,001	0.9984
S4	Loop rejection	6	0/1	0	120,001	0.0000
S5	Delayed restoration	46	2/2	5	920,001	39.9984
S6	Physical fault	7	2/2	5	140,001	1.1948

5 Experimental Method and Evidence Handling

5.1 Saved Scenario Matrix

The six electrical experiments were rerun against the integrated repository on 25 September 2026. Raw outputs were archived per condition outside the manuscript package, while this paper freezes their result summaries and figure previews. There is one saved deterministic result per condition in this refresh, so no confidence interval, stochastic reliability estimate, or repeated-run performance distribution is reported.

The six conditions are: S1, no-fault baseline; S2, N03 isolation at 5 s without restoration; S3, the same isolation followed by tie closure at 6 s; S4, attempted closure of a tie that would create a loop in the initial topology; S5, isolation at 5 s and deliberately delayed tie closure at 45 s; and S6, a native fault at 4.8–5.2 s with isolation at 5 s and restoration at 6 s. S2 and S3 retain “fault” terminology in some originating filenames or scenario descriptions, but contain only logical isolation commands, not an applied physical fault. Only S6 energizes the native fault component.

The input reason string in S6 refers to protection detecting a fault. This is scenario annotation, not evidence of a simulated detector. Both fault timing and controller timing are provided in advance in the same experiment input. The 0.2-s interval between fault inception and isolation is imposed; it does not measure relay decision time or communication delay.

5.2 Measured Channels and Interruption Metric

Each run provides six bus RMS voltage channels in kV; seven branch RMS-current channels in kA; seven branch active-power channels in MW; seven breaker-control state channels; three instantaneous fault-current channels in kA; and one fault-state channel. These sum to 31 research channels. The binary **State_*** channels record scheduled control signals, not independent mechanical-contact feedback. The fault-state signal uses 1 for active, unlike the breaker convention in Equation 3.

The raw output also contains animation-related traces. The existing reader distinguishes the research recorders from sparsely sampled display channels instead of treating every exported trace as a full-rate measurement. The saved integration evidence reports 146 raw traces and 31 selected research traces.

For recorded bus RMS voltage $V_i[n]$, define a threshold indicator

$$z_i[n] = \begin{cases} 1, & t_n \geq 0.5 \text{ s and } V_i[n] < 0.2 \text{ kV}, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

An interval begins at the first below-threshold sample and ends at the first sample no longer below threshold. The implemented analysis merges consecutive intervals separated by at most 0.02 s, then discards intervals shorter than 0.01 s. If an interval continues to the last sample, the saved duration is right-censored at the run end. For each retained interval j ,

$$D_{i,j} = t_{i,j}^{\text{end}} - t_{i,j}^{\text{start}}. \quad (7)$$

Because of the merge rule, $D_{i,j}$ can include very short recovered gaps; it is a sustained-interruption envelope, not necessarily the exact integral of z_i . The threshold is 0.5 per unit of the nominal 400-V voltage, chosen for this experiment. It is not a claim of compliance with a voltage-quality standard, nor a customer reliability index such as SAIDI.

The peak absolute current for fault phase ϕ is

$$I_{\phi}^{\text{pk}} = \max_n |i_{\phi}[n]|. \quad (8)$$

The pipeline computes peaks and threshold intervals from the full recorded series before reducing the displayed traces. Sample alignment is $50 \mu\text{s}$; this spacing is not a calibrated accuracy bound on RMS filter dynamics or switching-device behavior.

5.3 Provenance and Visualization

The source result file stores final values, extrema, event-aligned samples, interruption metrics, and a 501-point evenly indexed preview per channel. The manuscript script verifies channel counts, sample counts, preview order, finite values, final-value consistency, and interval arithmetic against this snapshot. It does not reconstruct full-rate waveforms from those previews. The numerical tables use stored full-rate summaries; the plotted voltage and RMS-current trajectories use explicitly labeled decimated previews.

In particular, evenly subsampled instantaneous fault current can alias the 50-Hz waveform and miss a peak. The paper therefore presents fault-current peak bars from the saved full-rate metrics rather than inventing a detailed waveform between preview points. Publication-grade transient waveform figures will require separately archived raw outputs.

Provenance includes the merged upstream revision, SHA-256 hashes of the read input artifacts and relevant source modules, the electrical build report, and the scenario inputs. A recorded graphical-validation artifact supplies additional evidence for baseline selection, physical-fault selection, and a command edit through the dashboard.

The merged-simulator integration case is separate from the six-row matrix. An N03 short-circuit-suspected event and an N05 communication-loss event are injected in the merged networking simulator. Its event objects pass through the domain-aware adapter; only the N03 event produces an isolation command. A native N03 PSCAD fault is preprogrammed at 4.8–5.2 s and isolation at 5.0 s. The networking simulator does not detect a PSCAD fault in a shared live run.

6 Results

6.1 Baseline, Isolation, and Loop Rejection

All six newly rerun matrix cases report 31 research channels, zero build errors, and zero build warnings. The recorded sample counts match their respective simulation durations at the specified output interval (Table 4). This is evidence of successful case execution and recorder extraction, not proof that the parameter set is physically calibrated.

In S1, the downstream final RMS voltages range from approximately 382.0 to 386.97 V at N03–N06. The normally open tie currents are negligible, and no retained below-threshold interval is reported. S2 opens the three incident branches of N03 at 5 s. N03–N06 then remain below the analysis threshold to the 6-s run end. N04 first crosses below 200 V at 5.0165 s, giving an observed duration of 0.9835 s. This is only a lower bound on the unresolved interruption, not a recovery time.

S4 rejects the single loop-forming closure and produces no state-changing switching events. Its final voltage profile remains consistent with the baseline. This establishes the intended behavior for that particular cycle test. It does not establish atomic rejection for a mixed command containing both acceptable and unacceptable operations.

6.2 Tie Restoration and Imposed Delay

In S3, tie closure at 6 s restores N04, N05, and N06 while N03 stays isolated. The respective final voltages are 380.51, 387.37, and 383.43 V (Table 5). The final tie currents are 32.54 A on N02–N05 and 13.45 A on N04–N06. These current magnitudes provide electrical evidence that the alternate branches carry load after closure, beyond a graphical indication that an edge is closed. They do not establish a thermal margin against measured cable ratings.

N04 and N06 remain below 200 V for approximately 0.9984 s in S3, and N05 for 0.99805 s. In S5, changing the tie-closure time from 6 to 45 s increases each of these saved interval durations by 39 s; their

Table 5: Saved final bus RMS voltage (V). The small N03 residual after restoration is not usable service; finite off-state paths remain in the model.

Bus	S1 baseline	S3 restoration	S6 physical fault
N03	386.97	0.02	0.02
N04	384.83	380.51	380.51
N05	383.51	387.37	387.37
N06	382.03	383.43	383.43

Table 6: Completed downstream intervals below 200 V, from saved full-rate metrics. Displayed decimal places reflect sample-based accounting rather than physical calibration accuracy.

Bus	S3 duration (s)	S5 duration (s)	S6 duration (s)
N04	0.99840	39.99840	1.19485
N05	0.99805	39.99805	1.19455
N06	0.99840	39.99840	1.19495

final voltage values remain the same to the displayed precision. The difference is the direct consequence of the imposed restoration schedule. It cannot be described as an SDN speedup over a measured legacy network, because neither network-delay distribution nor communications implementation is varied in these runs.

6.3 Native Physical Fault and Restoration

The physical-fault scenario produces a different voltage trajectory from isolation-only switching. The fault input is active from 4.8 to 5.2 s; N03–N06 cross below threshold around 4.820 s, before the scheduled breaker opening at 5 s. The maximum absolute fault phase currents are 1.552, 1.554, and 1.552 kA for A, B, and C, respectively. The largest recorded phase peak is 1.554333 kA. In contrast, inactive-fault cases exhibit approximately 1.3 mA maximum leakage rather than a high-current event.

Opening the three N03 incident branches removes the source-fed route to the fault location. The fault logic clears at 5.2 s, and the healthy downstream buses are reconnected through the two ties at 6 s. Their below-threshold durations are 1.19485 s at N04, 1.19455 s at N05, and 1.19495 s at N06. Their final voltages and tie currents agree with S3 at the reported precision. N03 remains isolated to the 7-s run end, with only a small residual measured voltage. Restoring healthy loads therefore does not imply restoration of the faulted bus.

Figure 3 combines the stored RMS trajectories with the exact scenario input schedule. Figure 4 reports the interval durations and fault-current peaks from the stored full-rate metrics. The roughly 15–20 ms difference between a programmed transition and an RMS threshold crossing should not be interpreted as communication latency. It includes the voltage measurement’s response and the electrical transient, as well as the threshold definition.

6.4 Merged Simulator and Dashboard Integration Evidence

The additional merged-simulator replay carries two generated network fault objects with explicit domains. The adapter emits one N03 power-isolation command, three breaker transitions, and no N05 communication-fault breaker command. A fresh 6.2-s PSCAD experiment records 124,001 samples on each of 31 channels, reports nine applied parameter writes, and returns no execution error. The native fault was supplied separately in advance, so this case verifies command compatibility and physical switching without claiming fault-triggered closed-loop protection.

The refreshed graphical-validation artifact dated 25 September 2026 records three successful actions: selecting the baseline, selecting the physical fault, and editing a controller command. All three retain the same PSCAD process identifier and return 31 selected channels with no reported errors. The baseline has 20,001 samples per channel; both fault-related runs have 140,001. The baseline fault-current peak is approximately 1.26×10^{-6} kA, while the phase-A peak for the physical-fault runs is approximately 1.552 kA. The command-edit validation moves isolation from 5.0 to 5.1 s while retaining the 6.0-s restoration

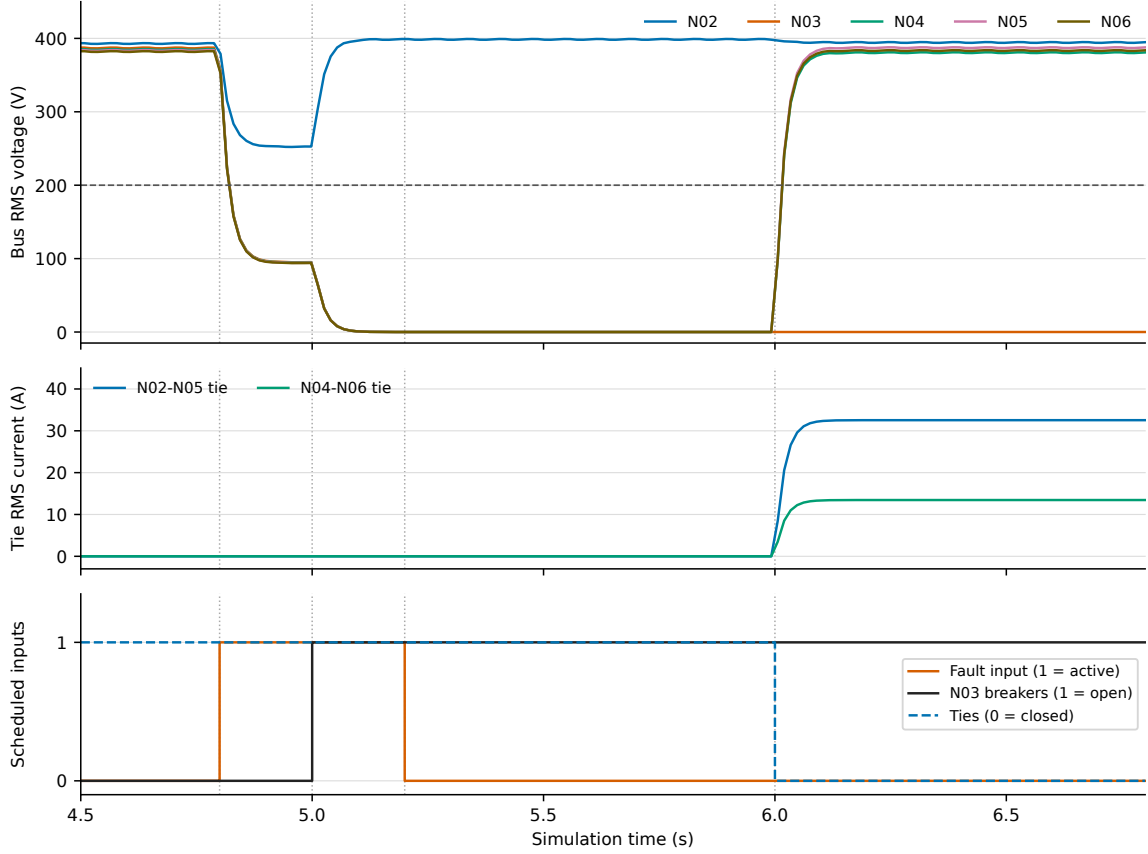


Figure 3: S6 physical-fault response near the switching events. The upper two panels use the stored 501-point voltage and RMS-current previews from a 140,001-sample-per-channel run; they are decimated trajectories, not full-rate transient waveforms. The lower panel shows exact input schedules, not measured breaker-contact feedback. Dotted lines mark 4.8, 5.0, 5.2, and 6.0 s.

time and the same fixed feeder model.

Together with the parameter manifests and fresh-output checks, this evidence supports the implemented dashboard-to-PSCAD batch workflow. It is not a measurement of end-to-end dashboard latency, proof of uninterrupted control connectivity, or a live feedback demonstration. The merged Python test suite passed 318 tests after the adapter update, including the compiled packet-codec checks. These software tests are not counted as electrical experiments.

7 Discussion, Limitations, and Research Agenda

7.1 What the Evidence Establishes

The saved artifacts show consistency across three representations: a requested network action, the accepted operation-level schedule, and the resulting electrical measurements. The isolation-only and physical-fault cases separate two phenomena that a purely graphical demonstration might conflate. Disconnecting a bus is not itself a short circuit; the native fault case adds an independently scheduled conductive path and measured fault current. Similarly, tie closure is supported by measured branch current and downstream voltage recovery, not just a changed line color.

The workflow is useful as a collaboration boundary. Networking colleagues can provide candidate routes without becoming responsible for PSCAD component identifiers, while electrical-model development can proceed without rewriting their routing code. The operation and measurement records make disagreements inspectable: a missing graph path, rejected closure, invalid mapping, failed build, stale output, and an electrically poor result are different failure classes. This separation supports debugging and subsequent experimental design.

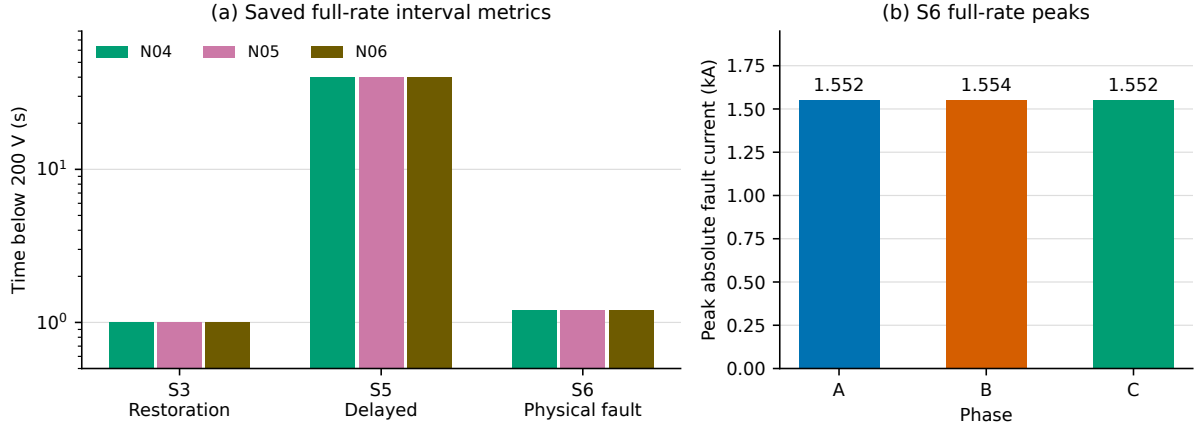


Figure 4: Saved full-rate summary metrics. Panel (a) uses a logarithmic duration axis to show the intentionally imposed long wait in S5 alongside S3 and S6. Panel (b) reports phase-current peaks for S6; no peaks are estimated from decimated waveform previews.

Nevertheless, traceability is not equivalent to electrical admissibility. Radiality and source connectivity are necessary modeling checks for the chosen scenario, but they are not a complete operating envelope. An overload or unacceptable voltage may occur in an acyclic, source-connected network. A future controller must use calibrated electrical constraints and independent interlocks before these commands could be considered for hardware.

7.2 Threats to Validity

Electrical validity. The six-bus network is intentionally small and uses illustrative branch impedances, source settings, and resistive loads. The load-sizing scalar is not the load power factor, and demand is voltage dependent. There is no validated transformer, grounding arrangement, DER controller, switching arc, protection curve, or grid-forming island. The restored voltage range should not be generalized to a campus or community microgrid without a new parameter set and validation.

Control and safety semantics. The implemented route checks are local and operation based. Partial command execution is possible, and command warnings are not equivalent to a transactional abort. This is a material limitation for generalizing the loop-rejection result. Atomic validation, explicit safe intermediate states, independent permissives, and rollback or recovery policies should precede any live or hardware-facing controller integration.

Timing and communications. No independent communication failure is simulated in the electrical run. The control delay is an input, not an observed transport characteristic. An always-connected control-plane objective remains untested, including its power supply, redundancy, reconnection behavior, and failure isolation. The merged networking simulator can represent communication faults, learned routes, binary packets, and radio contention. In this study those processes were not clock-coupled to PSCAD, and no measured packet timing was used as a breaker command delay. The MCP service is a local automation mechanism, not a safety-rated field protocol or evidence of secure, reliable network delivery.

Measurement and repeatability. The analysis threshold and interval filtering are experimental definitions. The selected breaker-state traces are controller outputs rather than independent contacts. The stored 501-point previews cannot support detailed fault-waveform reconstruction. No timestep-convergence study, independently recomputed per-scenario raw-output analysis, repeated-run variance, or instrument calibration is supplied with this draft. The saved deterministic matrix supports functional inspection, not statistical reliability claims.

Novelty and comparison. Software-defined microgrids and cyber-physical testbeds already exist. The current manuscript does not show that its automation architecture outperforms those systems, or that

the networking algorithm is optimal. A future venue submission must align its contribution with an appropriately scoped systems or experimental-methods claim, or add a genuinely new and evaluated control method.

7.3 Next Experimental Stage

The next stage should first freeze measured or benchmark electrical parameters, validate the steady state and fault level independently, and repeat the cases at multiple EMT time steps. A larger feeder and unbalanced loads would test whether the adapter remains useful beyond the current balanced six-bus example. Branch ampacity, source limits, neutral and grounding models, and load dynamics should be explicit inputs rather than inferred from network-capacity labels.

Live integration then needs an agreed controller transport with versioned messages, identifiers, acknowledgments, stale-command handling, and a well-defined failure policy. Fault detection must originate from measured electrical signals if the experiment is to evaluate protection response. Closing the measurement-to-controller loop also requires an explicit clock and synchronization strategy; established federation approaches such as HELICS and FNCS provide relevant design precedents [8, 7], but choosing or integrating one is future work.

A useful comparative protocol would hold the electrical case fixed and vary only the controller and communication treatment. Candidate conditions are manual or fixed restoration, the current supplied-route policy, and the networking team’s updated policy. Controlled delay, loss, disconnection, and recovery distributions should use documented seeds and repeated trials. Record detection time, issue time, receipt time, breaker operation, and load recovery separately; do not collapse them into a single “SDN delay.”

The evaluation should report operation-level and transaction-level rejection, voltage-limit violations, loading limits, switching count, recovery duration, and unavailable demand. For example, a future energy-not-served measure could be defined using a specified counterfactual demand trajectory:

$$E_{\text{NS}} = \sum_i \int \max\{P_i^{\text{required}}(t) - P_i^{\text{served}}(t), 0\} dt. \quad (9)$$

This metric is not computed here. In a resistive-load model, reduced consumed power during undervoltage is not automatically equivalent to disconnected customer demand; the counterfactual must be stated before calculating such an index. Comparative results should include uncertainty appropriate to the randomized treatments and retain all raw electrical and communication logs.

8 Conclusion

This paper describes a working batch interface from structured network-control commands and graphical dashboard edits to timed PSCAD electrical experiments. The modular implementation preserves the networking code boundary while linking logical identifiers, operation-level decisions, component parameters, fresh outputs, and measured results. Six freshly rerun deterministic scenarios provide preliminary functional evidence for baseline operation, isolation, tie-based restoration, single-loop rejection, imposed waiting time, and a native three-phase fault.

An additional event replay uses fault objects from the merged networking simulator. It produces electrical isolation for a power fault and leaves the switch schedule unchanged for a communication fault, confirming the fault-domain boundary under test.

In the physical-fault case, a maximum absolute phase current of 1.554 kA precedes restoration of healthy downstream buses to approximately 381–387 V, with about 1.195 s below the selected 200-V threshold. The faulted bus stays isolated. These observations support the command-to-EMT workflow, not a claim of optimal routing, measured SDN latency, or completed live microgrid protection. Calibrated models, atomic command semantics, independent fault detection, and synchronized controller feedback are the principal requirements for the next research stage.

Data and Code Availability

The manuscript package contains a frozen extract of the rerun six-scenario results, scenario inputs, electrical build metadata, provenance hashes, generated figures, and the scripts used to produce its tables

and plots. It does not contain a complete per-scenario archive of raw PSCAD output. The originating networking repository is <https://github.com/MathishaBuwaneke/SEIAN-AMRP-Mesh-Simulator>. That URL identifies the upstream networking work; it is not a claim that the local power-plane extension or this manuscript has been publicly released there. The availability and licensing of a submission-time research release remain to be confirmed by the authors.

Author and Funding Statements

Author order, affiliations, individual contributions, acknowledgments, funding, and competing-interest declarations must be completed by the project team before submission. No external funding or deployment authorization is asserted in this draft.

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A Reproducibility and Artifact Guide

The independent manuscript package generates its numerical tables and four original figures from a frozen saved-result snapshot. Its generator neither launches PSCAD nor writes into operational folders. Snapshot refresh is an explicit, read-only access to the source project.

Table 7: Manuscript-package contents and their evidential role.

Artifact	Role
<code>data/evidence_snapshot.json</code>	Saved scenario results, input batches, electrical metadata, and GUI-validation summary
<code>data/provenance.json</code>	Original input hashes, snapshot time, and inspected repository revision
<code>data/scenario_summary.csv</code>	Numeric values underlying the six-scenario table
<code>scripts/prepare_artifacts.py</code>	Read-only evidence extraction and figure/table generation
<code>figures/, tables/</code>	Generated manuscript assets
<code>research_notes.md</code>	Literature source audit, claim boundaries, and author checklist
<code>build.ps1</code>	Local LaTeX and BibTeX build into the paper’s build directory

The merged upstream repository revision is `4a6f49d`. Local adapter and validation changes made for this refresh are identified by SHA-256 hashes in the provenance file; the upstream revision alone does not identify them. Individual hashes identify the actual inputs used by the manuscript. The existing project context records PSCAD 5.0.1 Educational, Python 3.13.7, `mhi.pscad` 2.9.7, `mhi.psout` 1.3.0, PowerMCP 0.3.0, MCP 2.1.1, Streamlit 1.63.0, and Playwright 1.62.0. These are recorded local versions, not a complete environment lock or a claim of compatibility across other installations. PSCAD licensing and a compatible compiler remain requirements for independently rerunning the electrical experiments.

The source workspace is `research_power_plane/pscad_workspace`. Its case and build metadata are saved there. The refreshed matrix, GUI evidence, merged-simulator replay, and separate raw outputs are under `output/pscad_refresh_20260925` in the working checkout; these files are ignored by Git, while the paper contains a compact hashed snapshot. Source modules governing the principal claims are `power_plane.py` (per-operation acceptance), `timeline.py` (state-changing schedules), `pscad_adapter.py` (component mapping), `pscad_mcp_client.py` and `pscad_mcp_server.py` (execution), and `transient_analysis.py` (metric definitions).

Rebuild the paper. Run the artifact generator and then the PDF build from this folder:

```
py -B scripts/prepare_artifacts.py
.\build.ps1
```

Only artifact generation needs Python, Matplotlib, and NumPy. The document uses the conventional `article` class and BibTeX; no online service or venue-specific template is required.

Reproduce the science. The per-scenario raw outputs from this refresh are archived under `output/pscad_refresh_20260925/raw` in the working checkout. The paper package should be distributed with a stable raw-output archive for independent reproduction. Rebuilding tables does not reproduce the EMT experiments. A scientific release must additionally archive the exact case and component library, compiler and dependency versions, initialization, command and fault manifests, per-scenario `.psout` files, solver messages, and rerunnable analysis. Archive each raw output before another run replaces it. Author and venue review is necessary before public release.